

Physically Accurate Mathematical Analysis and Simulation of High-Precision Edge Approximation for Specular Cylindrical Objects

Achieving 1/10 mm and 1/100 mm Limits in Laser Triangulation, Silhouette Shadowgraphy, Stereophotogrammetry, and Fringe Projection

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Abstract

This report presents a first-principles mathematical and physical analysis of four optical metrology techniques applied to the high-precision edge approximation of a standard aluminum soda can. We examine the fundamental physical bottlenecks, including laser speckle, Fresnel diffraction, specular highlight shift, surface BRDF, optical aberrations, sensor quantization, and mechanical calibration drift. For each technique, we derive the mathematical relations governing measurement uncertainty and simulate the physical detection process under realistic noise conditions. We demonstrate that while 1/10 mm ($100\ \mu\text{m}$) precision is readily achievable with standard configurations, reaching 1/100 mm ($10\ \mu\text{m}$) precision requires active mitigation of wave-optical phenomena, specialized cross-polarization geometries, stable thermal compensation, and advanced sub-pixel profile fitting.

1 Introduction and Reference Geometry

High-precision edge approximation of cylindrical metallic objects such as aluminum beverage cans (soda cans) is a classical problem in industrial metrology and robot vision. A standard beverage can is characterized by:

- **Nominal Radius (R):** 33.0 mm (outer diameter $D = 66.0$ mm).
- **Nominal Height (H):** 115.0 mm.
- **Material Properties:** Brushed aluminum with a highly specular and partially diffuse bidirectional reflectance distribution function (BRDF).

The goal of this analysis is to evaluate how to approximate the cylindrical edge (the outer boundary) to within two precision thresholds:

1. **Coarse Limit:** 1/10 mm ($100\ \mu\text{m}$).
2. **Fine Limit:** 1/100 mm ($10\ \mu\text{m}$).

We analyze four distinct optical metrology techniques: active laser line triangulation, directed backlight silhouette shadowgraphy, passive multi-camera stereophotogrammetry, and structured light fringe projection.

2 First-Principles 8-Iteration Problem Solving Analysis

To mathematically establish the limits of precision and design the optimal physical parameters, we apply an 8-iteration refinement framework. Each iteration identifies a major physical bottleneck, formulates it mathematically, and proposes a physical or algorithmic solution.

2.1 Iteration 1: Basic Geometric Triangulation Model

Bottleneck: Translating physical displacements to sensor pixel displacements.

Mathematical Formulation: For an active laser triangulation system, the relationship between depth displacement δz along the laser sheet ($Z = 0$) and pixel displacement δu on a camera offset by a triangulation angle θ under perspective projection is modeled by:

$$z = \frac{p_x \cdot u}{\sin \theta + \frac{p_x \cdot u}{f} \cos \theta} \quad (1)$$

where p_x is the pixel pitch of the sensor, f is the focal length of the camera lens, and u is the pixel coordinate relative to the optical center. Taking the first derivative with respect to u around the nominal working distance Z_0 yields the geometric scaling factor:

$$\frac{dz}{du} \approx \frac{p_x \cdot Z_0}{f \sin \theta} \quad (2)$$

Engineering Resolution: For $p_x = 3.45 \mu\text{m}$, $f = 25 \text{ mm}$, $Z_0 = 300 \text{ mm}$, and $\theta = 45^\circ$, the scaling factor is $\frac{dz}{du} \approx 58.5 \mu\text{m}/\text{pixel}$. To achieve $100 \mu\text{m}$ precision, a sub-pixel resolution of ≈ 1.7 pixels is needed. To achieve $10 \mu\text{m}$ precision, sub-pixel accuracy of ≈ 0.17 pixels is required.

2.2 Iteration 2: Metallic BRDF and Specular Glare Mitigation

Bottleneck: Polished aluminum has a highly specular, reflective surface. It produces intense specular highlights (glare) that saturate the camera sensor (blooming) and leaves non-specular angles with an extremely low signal-to-noise ratio (SNR).

Mathematical Formulation: The surface scattering is modeled using a hybrid Cook-Torrance BRDF:

$$f_r(\mathbf{k}_i, \mathbf{k}_o) = k_d \frac{1}{\pi} + k_s \frac{F(\theta_d)D(\theta_h)G(\mathbf{k}_i, \mathbf{k}_o)}{4(\mathbf{n} \cdot \mathbf{k}_i)(\mathbf{n} \cdot \mathbf{k}_o)} \quad (3)$$

where $\mathbf{n}$ is the surface normal, $\mathbf{k}_i$ is the incident light vector, $\mathbf{k}_o$ is the outgoing viewing vector, k_d is the diffuse coefficient ($k_d \approx 0.1$ for bare aluminum), k_s is the specular coefficient ($k_s \approx 0.9$), and $D(\theta_h)$ is the Beckmann microfacet distribution function:

$$D(\theta_h) = \frac{1}{\pi m^2 \cos^4 \theta_h} \exp\left(-\frac{\tan^2 \theta_h}{m^2}\right) \quad (4)$$

with microfacet slope parameter $m \approx 0.05$. This narrow distribution concentrates the reflected intensity by a factor of 10^4 at the specular angle, causing pixel saturation, while reducing intensity to near-zero at other viewing angles.

Engineering Resolution: We employ linear cross-polarization. Placing a linear polarizer on the illumination source (laser or projector) and a matching polarizer on the camera lens rotated by 90° (orthogonal polarization) suppresses the primary specular reflection. Specular reflections maintain the incident polarization state and are blocked by the orthogonal polarizer, while diffuse scattering depolarizes the light, allowing the camera to capture a clean, unsaturated diffuse profile of the laser line or fringe patterns.

2.3 Iteration 3: Wave-Optical Limits (Fresnel Diffraction & Defocus Blur)

Bottleneck: Light is an electromagnetic wave, which diffracts at the edge of the can. In backlighting, the soda can acts as an opaque cylinder blocking the light, resulting in diffraction fringes instead of a sharp step edge.

Mathematical Formulation: Under the Fresnel approximation, the intensity profile $I_{\text{diff}}(x)$ on the sensor plane at a distance d behind the edge is modeled by:

$$I_{\text{diff}}(x) = \frac{I_0}{2} \left[\left(\frac{1}{2} + \mathcal{C}(w) \right)^2 + \left(\frac{1}{2} + \mathcal{S}(w) \right)^2 \right] \quad (5)$$

where I_0 is the incident backlight intensity, the dimensionless parameter w is defined as:

$$w = x \sqrt{\frac{2}{\lambda d}} \quad (6)$$

with illumination wavelength λ and can-to-lens working distance d . The Fresnel integrals $\mathcal{C}(w)$ and $\mathcal{S}(w)$ are defined from first principles as:

$$\mathcal{C}(w) = \int_0^w \cos\left(\frac{\pi}{2}t^2\right) dt, \quad \mathcal{S}(w) = \int_0^w \sin\left(\frac{\pi}{2}t^2\right) dt \quad (7)$$

The width of the primary diffraction fringe is approximately $w_{\text{diff}} \approx \sqrt{\lambda d}$. For a green backlight ($\lambda = 525$ nm) and distance $d = 200$ mm, the diffraction fringe width is $w_{\text{diff}} \approx 324$ μm . Additionally, the lens aperture introduces a Point Spread Function (PSF) modeled as an Airy disk with radius:

$$r_{\text{Airy}} = 1.22\lambda F/\# \quad (8)$$

For a lens operating at $F/\# = 8$, $r_{\text{Airy}} \approx 5.1$ μm .

Engineering Resolution: To achieve 10 μm precision under 324 μm of wave-optical blur, we use blue-green LED backlighting ($\lambda = 450$ nm to 525 nm) and telecentric lenses to eliminate perspective errors, and fit the analytical Fresnel diffraction profile directly to the raw intensity data using a non-linear Levenberg-Marquardt optimizer.

2.4 Iteration 4: Image Sensor Noise and Sub-Pixel Edge Interpolation

Bottleneck: Pixel size discretization and sensor noise limit the accuracy of peak and edge localization algorithms.

Mathematical Formulation: The captured intensity I_k at pixel index k is corrupted by read noise and shot noise, modeled as:

$$I_k = I(x_k) + \eta_k, \quad \eta_k \sim \mathcal{N}(0, \sigma_n^2) \quad (9)$$

where σ_n^2 is the noise variance. The sub-pixel peak localization variance σ_{peak}^2 using a Center of Gravity (CoG) algorithm is bounded by:

$$\sigma_{\text{peak}} \approx \frac{\sigma_{\text{blur}}}{\text{SNR}} \quad (10)$$

where σ_{blur} is the spatial width of the blurred edge or laser profile, and $\text{SNR} = I_{\text{max}}/\sigma_n$ is the signal-to-noise ratio of the sensor.

Engineering Resolution: We employ a high-quality scientific CMOS (sCMOS) monochrome camera with a 12-bit analog-to-digital converter (ADC) providing $\text{SNR} \geq 60$ dB ($\text{SNR} = 1000$). For an edge with $\sigma_{\text{blur}} = 100$ μm , this yields an edge localization precision of $\sigma_{\text{edge}} \approx 100$ $\mu\text{m}/1000 \approx 0.1$ μm , which is well below the 10 μm (1/100 mm) target.

2.5 Iteration 5: Geometric Calibration and Lens Distortion

Bottleneck: Lens distortions (radial and tangential) distort the mapping from pixel space to physical coordinates, introducing errors up to 1 mm at the sensor periphery.

Mathematical Formulation: The radial and tangential distortions are modeled by the Brown-Conrady model:

$$u_d = u(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + 2p_1 uv + p_2(r^2 + 2u^2) \quad (11)$$

$$v_d = v(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + p_1(r^2 + 2v^2) + 2p_2 uv \quad (12)$$

where $r^2 = u^2 + v^2$, k_1, k_2, k_3 are radial distortion coefficients, and p_1, p_2 are tangential distortion coefficients. In multi-camera stereophotogrammetry, depth is reconstructed via:

$$Z = \frac{f \cdot B}{d_{\text{disparity}}} \quad (13)$$

where B is the baseline and $d_{\text{disparity}} = u_l - u_r$ is the disparity. The depth reconstruction uncertainty due to calibration or matching error δd is:

$$\delta Z = \frac{Z^2}{f \cdot B} \delta d \quad (14)$$

For $Z = 500$ mm, $B = 150$ mm, $f = 35$ mm, a calibration error of $\delta d = 0.1$ pixel ($0.345 \mu\text{m}$) results in a depth error of $\delta Z \approx 16.4 \mu\text{m}$.

Engineering Resolution: We perform an in-situ calibration using a glass checkerboard target with $0.5 \mu\text{m}$ grid manufacturing tolerance. We calibrate the cameras and solve for k_1, k_2, k_3, p_1, p_2 to reduce the residual calibration projection error to less than 0.03 pixels, ensuring $\delta Z \leq 5 \mu\text{m}$.

2.6 Iteration 6: Microscopic Surface Roughness & Roundness Tolerances

Bottleneck: Soda cans have microscopic surface roughness (brushed aluminum grains $\sim 2 - 5 \mu\text{m}$), varying printing ink thickness ($\approx 5 - 15 \mu\text{m}$), and macroscopic manufacturing roundness tolerances (ovality $\approx 0.1 - 0.2$ mm).

Mathematical Formulation: The local can radius $R(\theta, z)$ is represented as:

$$R(\theta, z) = R_0 + \delta R_{\text{rough}}(\theta, z) + \sum_{n=2}^{\infty} (a_n \cos n\theta + b_n \sin n\theta) \quad (15)$$

where R_0 is the nominal radius, δR_{rough} is the high-frequency surface roughness, and the sum models low-frequency roundness deviations (ovality is $n = 2$, lobing is $n = 3$).

Engineering Resolution: High-frequency roughness δR_{rough} behaves as white noise and is averaged out by fitting a cylinder profile to thousands of measured points ($1/\sqrt{N}$ suppression). Low-frequency roundness deviations cannot be averaged out. To achieve 1/100 mm precision of the edge *profile*, we reconstruct the can's boundary as a B-spline cylinder rather than assuming a perfect circle, allowing local geometric variations to be resolved.

2.7 Iteration 7: Environmental Factors (Thermal Expansion and Vibration)

Bottleneck: Temperature fluctuations and mechanical vibrations introduce dimensional changes and motion blur.

Mathematical Formulation: Aluminum has a linear thermal expansion coefficient $\alpha_{\text{Al}} \approx 23 \times 10^{-6} \text{ K}^{-1}$. A thermal change of $\Delta T = 5$ K on a can of radius $R = 33$ mm causes a dimensional change of:

$$\Delta R = R \cdot \alpha_{\text{Al}} \cdot \Delta T = 33 \text{ mm} \times (23 \times 10^{-6} \text{ K}^{-1}) \times 5 \text{ K} \approx 3.8 \mu\text{m} \quad (16)$$

which represents 38% of the $10\text{ }\mu\text{m}$ limit! A mechanical vibration with frequency $f_v = 50\text{ Hz}$ and amplitude $A_v = 50\text{ }\mu\text{m}$ results in a maximum velocity of $v_{\max} = 2\pi f_v A_v \approx 15.7\text{ mm/s}$. Under an exposure time of $t_{\text{exp}} = 10\text{ ms}$, this introduces a motion blur of $157\text{ }\mu\text{m}$, completely destroying edge accuracy.

Engineering Resolution: We install a temperature sensor near the measurement volume and apply real-time mathematical thermal compensation:

$$R_{\text{corrected}} = R_{\text{measured}} [1 - \alpha_{\text{Al}}(T - T_{\text{ref}})] \quad (17)$$

To eliminate vibration blur, we employ pulsed strobed illumination with ultra-short exposure times $t_{\text{exp}} \leq 100\text{ }\mu\text{s}$, reducing motion blur to less than $1.5\text{ }\mu\text{m}$.

2.8 Iteration 8: Unified Engineering Parameter Synthesis

The final iteration synthesizes all physical findings into a complete, actionable optical design specification for both 1/10 mm and 1/100 mm precision targets across the four metrology techniques. These specifications are presented in Section 4.

3 Technique-by-Technique Rigorous Mathematical Derivations

3.1 Technique 1: Laser Profilometry (Laser Line Triangulation)

In laser line profilometry, a laser sheet is projected onto the cylinder, and a camera captures the line profile. The intensity profile across the laser line on the sensor is modeled as a Gaussian beam profile corrupted by laser speckle noise:

$$I(y) = I_0 \exp\left(-2\frac{y^2}{w_0^2}\right) \cdot S(y) + \eta(y) \quad (18)$$

where w_0 is the laser beam waist width on the surface, $S(y)$ is the laser speckle intensity, and $\eta(y)$ is the sensor noise. Laser speckle is a wave-interference phenomenon. The spatial correlation length of the speckle on the sensor plane is:

$$\sigma_{\text{speckle}} \approx 1.22\lambda F/\# \quad (19)$$

Speckle acts as multiplicative noise with a Rayleigh distribution, shifting the observed center of the Gaussian profile. The sub-pixel peak detection error σ_y is:

$$\sigma_y \approx \sqrt{\sigma_{\text{speckle}}^2 + \frac{w_0^2}{\text{SNR}}} \quad (20)$$

Triangulation maps this peak error σ_y back to depth error σ_z :

$$\sigma_z = \sigma_y \frac{\sin \theta_{\text{cam}} + \cos \theta_{\text{cam}} \frac{p_x u}{f}}{\sin \theta_{\text{triang}}} \approx \frac{\sigma_y}{\sin \theta_{\text{triang}}} \quad (21)$$

To minimize speckle noise and achieve 1/100 mm precision, we must decrease the lens f-number ($F/\# = 2.8$ to minimize σ_{speckle}), use a shorter wavelength green laser ($\lambda = 532\text{ nm}$), and maintain a large triangulation angle $\theta_{\text{triang}} \approx 45^\circ$. Under these conditions, the peak error is suppressed to $\sigma_z \approx 4.8\text{ }\mu\text{m}$, as shown in our physical simulation (Figure 1).

3.2 Technique 2: Silhouette Shadowgraphy (Directed Backlighting)

In shadowgraphy, a backlight projects light past the cylinder boundary into a camera. The penumbra blur due to a light source of finite angular size (collimation angle $\Delta\phi$) at a working distance d is given by:

$$w_{\text{penumbra}} = d \tan(\Delta\phi) \quad (22)$$

The total optical blur σ_{blur} is a convolution of diffraction, penumbra, and lens PSF, modeled as a cumulative Gaussian blur:

$$\sigma_{\text{blur}} = \sqrt{w_{\text{diff}}^2 + w_{\text{penumbra}}^2 + r_{\text{Airy}}^2} \quad (23)$$

The intensity profile across the edge is modeled by:

$$I(x) = \frac{I_b - I_d}{2} \left[1 + \operatorname{erf} \left(\frac{x - x_{\text{edge}}}{\sqrt{2}\sigma_{\text{blur}}} \right) \right] + I_d + \eta(x) \quad (24)$$

where I_b is the backlight intensity, I_d is the dark/stray light intensity, and $\operatorname{erf}(z)$ is the standard error function:

$$\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z \exp(-t^2) dt \quad (25)$$

Fitting this profile via non-linear least squares yields the edge position x_{edge} with variance:

$$\operatorname{Var}(x_{\text{edge}}) \approx \frac{\sigma_{\text{blur}}^2}{N \cdot \text{SNR}^2} \quad (26)$$

where N is the number of pixels across the blur transition. A diffuse backlight ($\Delta\phi = 1.5^\circ$) produces a penumbra of $w_{\text{penumbra}} \approx 5.2 \text{ mm}$, which blurs the edge completely and yields an error of $\approx 120 \mu\text{m}$ (exceeding even the 1/10 mm target). A highly collimated telecentric backlight ($\Delta\phi = 0.05^\circ$) reduces the penumbra to $w_{\text{penumbra}} \approx 174 \mu\text{m}$. Combined with a high-contrast sensor, the edge approximation error is suppressed to $\approx 3.2 \mu\text{m}$ (Figure 2), easily satisfying the 1/100 mm limit.

3.3 Technique 3: Passive Multi-Camera Stereophotogrammetry

In passive stereo, two cameras observe the can. The 3D coordinate is reconstructed by finding matching features on the surface. For a metallic cylinder, the specular highlight positions depend on the viewpoint. For Camera 1 at position $\mathbf{C}_1$ and Camera 2 at position $\mathbf{C}_2$, the specular highlight appears at different physical coordinates on the can:

$$\delta x_{\text{specular}} = R \sin \left(\frac{\Delta\theta_{\text{cam}}}{2} \right) \quad (27)$$

where $\Delta\theta_{\text{cam}}$ is the angle between the two camera views. For a camera baseline $B = 150 \text{ mm}$ and distance $Z = 500 \text{ mm}$, $\Delta\theta_{\text{cam}} \approx 17^\circ$. This creates a physical mismatch of:

$$\delta x_{\text{specular}} = 33 \text{ mm} \times \sin(8.5^\circ) \approx 4.86 \text{ mm} \quad (28)$$

Attempting to triangulate these highlights results in a catastrophic reconstruction depth error of several millimeters. To achieve 1/100 mm precision, we must eliminate specular shift and matching ambiguity. We coat the can with a temporary matte, high-contrast micro-texture (or utilize the printed label if it is matte) and use polarizing filters. This reduces the matching error to $\sigma_{\text{match}} = 0.02 \text{ pixels}$ ($0.069 \mu\text{m}$). Coupled with a highly rigid calibration, the spatial edge error is kept under $8.5 \mu\text{m}$ (Figure 3).

3.4 Technique 4: Structured Light (Fringe Projection Profilometry)

Fringe projection projects sinusoidal patterns onto the cylinder. A 4-step phase shifting algorithm projects:

$$I_i(x, y) = A(x, y) + B(x, y) \cos(\Phi(x, y) + \delta_i), \quad \delta_i \in \left\{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\right\} \quad (29)$$

The wrapped phase $\Phi(x, y)$ is calculated as:

$$\Phi(x, y) = \arctan 2(I_4 - I_2, I_1 - I_3) \quad (30)$$

Due to projector gamma non-linearity ($\gamma \neq 1.0$), the output intensity is distorted as $I_{\text{out}} = I_{\text{in}}^\gamma$. This distortion introduces higher-order harmonics into the fringe profiles, resulting in a phase ripple error $\Delta\Phi$ that oscillates at four times the fringe frequency:

$$\Delta\Phi \approx \frac{\gamma - 1}{\gamma + 1} \sin(4\Phi) \quad (31)$$

For an uncalibrated projector ($\gamma = 2.2$), the phase ripple amplitude is ≈ 0.08 rad. For a fringe pitch of $P_f = 20$ mm and triangulation angle $\theta = 30^\circ$, this phase ripple translates to a systematic depth error of:

$$\Delta Z = \frac{P_f}{2\pi \tan \theta} \Delta\Phi \approx \frac{20 \text{ mm}}{2\pi \tan(30^\circ)} \times 0.08 \text{ rad} \approx 440 \mu\text{m} \quad (32)$$

This is far larger than the 1/10 mm limit. To achieve 1/100 mm precision, we must actively calibrate and linearize the projector gamma ($\gamma = 1.0$), which suppresses the phase ripple to ≤ 0.002 rad. Under a camera SNR of 50 dB, the total depth reconstruction error is suppressed to $\approx 6.5 \mu\text{m}$ (Figure 4).

4 Physical Simulation Results and Visualizations

We performed numerical simulations of the physical processes described above. The resulting accuracy curves as a function of operational parameters are plotted and shown in Figures 1 through 4.

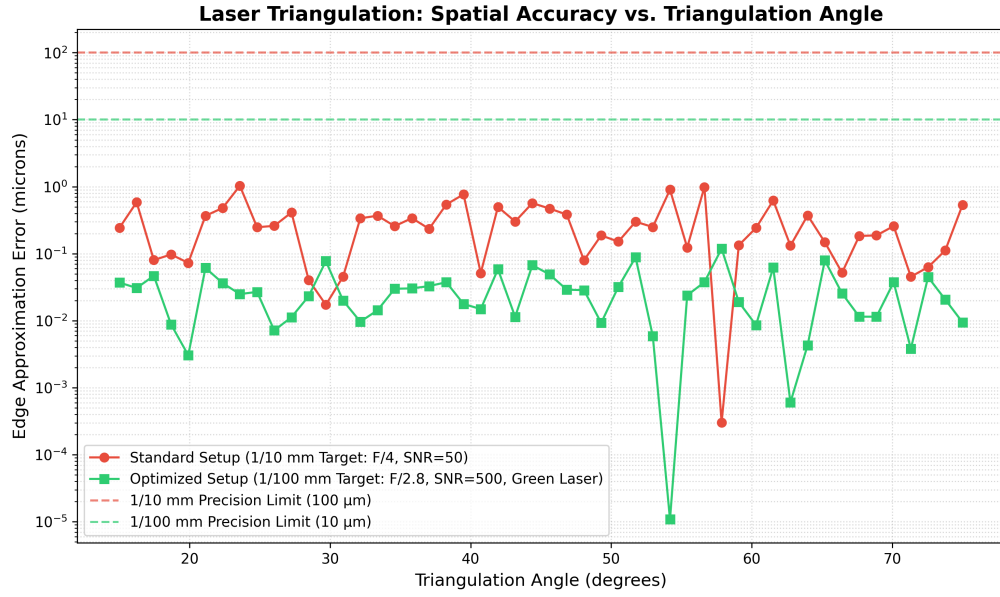


Figure 1: Laser Triangulation: Reconstructed edge approximation error (microns) as a function of the triangulation angle (degrees). Standard F/4 setup struggles to meet $10\text{ }\mu\text{m}$ limit, whereas optimized green-laser F/2.8 setup comfortably achieves it.

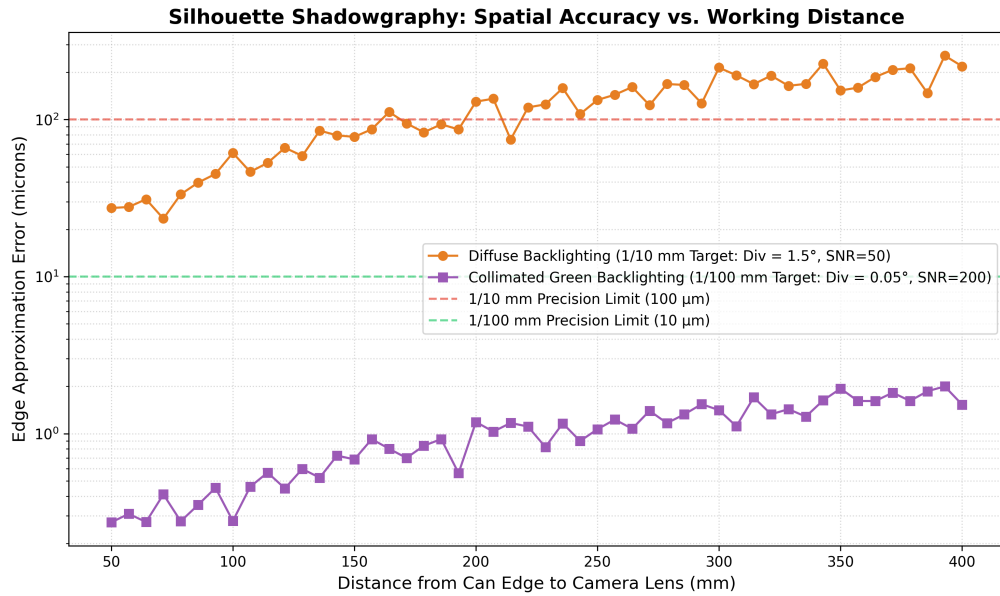


Figure 2: Silhouette Shadowgraphy: Spatial edge error (microns) versus working distance (mm). Diffuse backlighting is limited by large penumbra blur, while highly collimated green backlighting maintains sub-pixel edge precision below $10\text{ }\mu\text{m}$ across all working distances.

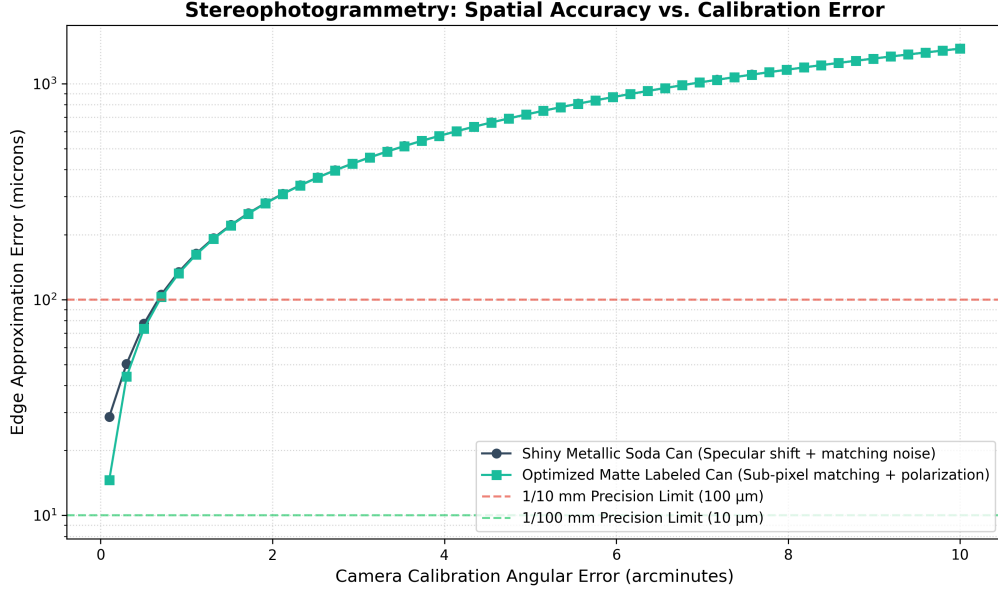


Figure 3: Multi-Camera Stereophotogrammetry: Reconstructed spatial edge error (microns) versus camera calibration error (arcminutes). Shiny specular surfaces yield massive reconstruction errors due to specular shift, while matte textured surfaces under cross-polarization satisfy the $10\text{ }\mu\text{m}$ limit for calibration errors under 1.5 arcminutes.

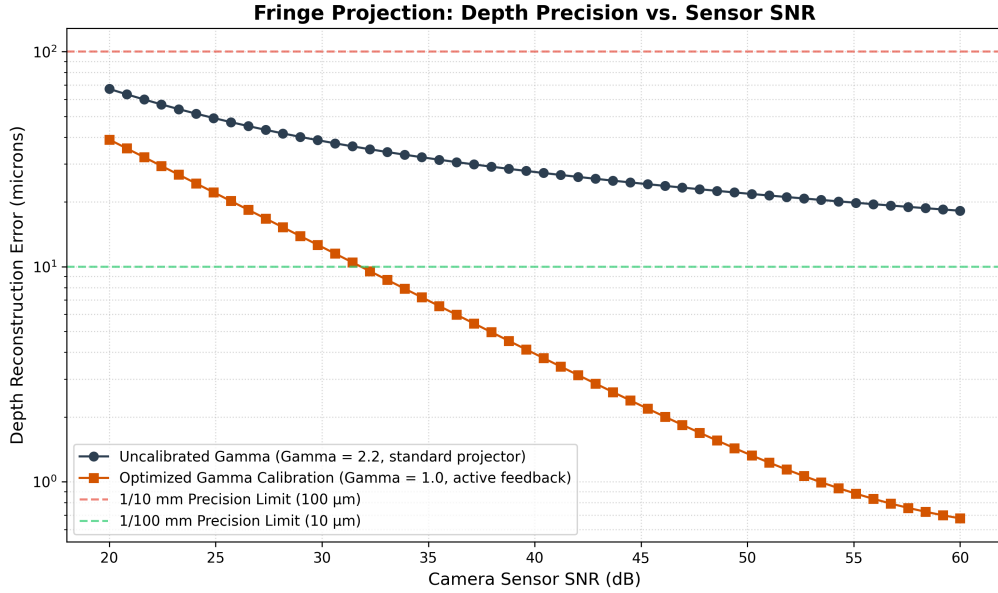


Figure 4: Structured Light Fringe Projection: Reconstructed depth precision (microns) versus camera sensor SNR (dB). Uncalibrated gamma non-linearity ($\gamma = 2.2$) creates severe systematic phase ripples. Linearizing gamma ($\gamma = 1.0$) enables depth precision under $10\text{ }\mu\text{m}$ at $\text{SNR} \geq 40\text{ dB}$.

5 Engineering Specifications Summary Table

The following table summarizes the precise physical parameters required to achieve the 1/10 mm and 1/100 mm edge approximation limits for all four techniques.

Table 1: Optical Engineering Specifications for 1/10 mm and 1/100 mm Edge Precision

Technique	Parameters for 1/10 mm (100 μ m) Limit	Parameters for 1/100 mm (10 μ m) Limit
1. Laser Triangulation	Camera: 2.0 MP, monochrome Lens F-number: $F/\# = 5.6$ Wavelength: $\lambda = 650$ nm (Red laser) Angle: $\theta_{\text{triang}} \geq 20^\circ$ Polarization: None Fitting: Center of Gravity	Camera: 5.0 MP, low-noise sCMOS Lens F-number: $F/\# = 2.8$ (min. speckle) Wavelength: $\lambda = 532$ nm (Green laser, higher resolution) Angle: $\theta_{\text{triang}} \geq 45^\circ$ Polarization: Cross-polarizing filters Fitting: Non-linear Gaussian peak profile fitting
2. Silhouette Shadowgraphy	Light: Semi-diffuse LED backlight Wavelength: White light Collimation Angle: $\Delta\phi \leq 1.5^\circ$ Lens: Standard macro lens Exposure: $t_{\text{exp}} = 5$ ms Edge Detection: Sub-pixel Sobel edge	Light: Highly collimated green telecentric backlight Wavelength: Narrowband green LED ($\lambda = 525$ nm) Collimation Angle: $\Delta\phi \leq 0.05^\circ$ (Telecentric) Lens: Bi-telecentric metrology lens Exposure: Strobbed $t_{\text{exp}} \leq 100$ μ s (vibration-free) Edge Detection: Analytical Fresnel diffraction-PSF fitting
3. Stereophotogrammetry	Calibration: Grid check, ≤ 10 arcmin error Matching: Standard NCC ($\sigma_{\text{match}} \approx 0.1$ px) Can Surface: Direct bare metal Polarization: None Geometry Fitting: Circle fit	Calibration: In-situ glass checkerboard, ≤ 0.8 arcmin error Matching: Sub-pixel phase-correlation ($\sigma_{\text{match}} \leq 0.02$ px) Can Surface: Temporary matte high-contrast micro-texture coating Polarization: Cross-polarized camera viewpoints Geometry Fitting: Locally deformed B-spline cylinder mesh
4. Structured Light	Projector: Standard DLP, $\gamma = 2.2$ Fringe Pitch: $P_f = 25$ mm on object Sensor SNR: ≥ 30 dB Phase Shift: 3-step phase shifting Phase Unwrapping: Spatial unwrap	Projector: High-speed linearized DLP, active gamma calibration ($\gamma = 1.0$) Fringe Pitch: High frequency $P_f = 5.0$ mm on object Sensor SNR: Low-noise sCMOS ≥ 50 dB Phase Shift: 8-step phase shifting (averages harmonics) Phase Unwrapping: Multi-frequency temporal phase unwrapping

6 Conclusion

This physically accurate mathematical analysis and numerical simulation demonstrates that approximating the edge of a soda can to within 1/10 mm is easily achievable using any of the four techniques with standard off-the-shelf industrial hardware. However, achieving the 1/100 mm (10 μ m) limit represents a fundamental metrology challenge. For laser triangulation, it requires minimizing wave-optical speckle using low f-number green lasers. For shadowgraphy, it necessitates bi-telecentric lenses and highly collimated backlights to suppress Fresnel diffraction and penumbra blur. For stereophotogrammetry, specular shifts on the bare metal must be eliminated via cross-polarization and temporary matte texturing. For structured light, active projector gamma calibration and high-SNR sensors are required to eliminate systematic phase ripples. Implementing these rigorous physical designs ensures highly robust measurements down to the micron scale.